

Actuator-Sensor-Placement in Dynamic Systems

**Mechatronic Systems Laboratory
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Contents

1	Introduction	3
2	Mode shape and operating deflection shape	3
2.1	Mode shape	3
2.2	Operating deflection shape (ODS)	3
3	Bending vibration of beams	4
3.1	Natural frequency and mode shape	4
3.2	Transfer function and operating deflection shape	7
4	Exercise Instruction	8
4.1	Preparation (Homework)	8
4.2	Precautions	8
4.3	Experiments	8
4.4	Hints and Recommendations	9
4.5	Report	9

1 Introduction

Understanding the dynamic behaviour of mechanical structures is essential for the design of mechatronic systems. The goals of this exercise are:

- Illustrate the difference between *mode shape* and *operating deflection shape (ODS)*
- Give insight into the dynamic response of bending beams
- Demonstrate, by means of a real experiment, that the controllability and observability of a dynamic system are dependent on the location of the actuator and the sensor

2 Mode shape and operating deflection shape

An accurate and relatively simple explanation of the difference between mode shapes and operating deflection shapes is given in [1].

2.1 Mode shape

A mode shape (or simply mode) is a specific motion pattern of a mechanical structure, occurring at a determinate frequency, called *resonance frequency*. Modes are inherent properties of the structure and depend only on the mass, stiffness and internal damping of the structure itself. They can be defined as the solution to the differential equation of motion:

$$[M]\{\ddot{x}(t)\} + [C]\{\dot{x}(t)\} + [K]\{x(t)\} = \{0\}, \quad (1)$$

where $[M]$, $[C]$ and $[K]$ are the mass, damping and stiffness matrices respectively and $\{x(t)\}$ is the displacement. Equation (1) can be written in the frequency domain as:

$$\{[M](j\omega)^2 + [C](j\omega) + [K]\}\{X(j\omega)\} = \{0\}. \quad (2)$$

Supposing that the matrices $[M]$, $[C]$ and $[K]$ have size n , Equation (2) will yield n solutions. Each of these solutions consists of the value of the resonance frequency $f_n = \omega_n/2\pi$ at which a specific mode occurs, together with the expression of the mode itself.

2.2 Operating deflection shape (ODS)

Differently from mode shapes, operating deflection shapes (or ODSs) depend not only on the inherent properties of mass, stiffness and damping of a structure but also on the external force $f(t)$ acting on the structure itself. An ODS is defined as *any forced motion of two or more points in a structure* [1] and is not necessarily frequency dependent. ODSs can be identified as the solutions to equation (3).

$$[M]\{\ddot{x}(t)\} + [C]\{\dot{x}(t)\} + [K]\{x(t)\} = \{f(t)\}. \quad (3)$$

From the previous explanation follows that *mode shapes can never be observed in nature*, as it is impossible to make a structure vibrate without applying an external excitation. Nonetheless, ODSs can be observed in nature and can be made to *approximate* a certain mode shape.

For this approximation to be acceptable, three conditions have to be met:

1. The excitation force $f(t)$ has to be applied on a position that *is not* a nodal line/nodal point for the mode shape to be approximated (i.e. a line/point where that mode shape is zero),
2. The excitation frequency has to be as close as possible to the resonance frequency of that specific mode shape,
3. At the resonance frequency, the ODS has to be dominated by one mode.

These conditions (especially condition 3.) are very hard to satisfy in real mechanical structures, given the geometric complication and the occurrence of multiple modes at close-by frequencies. One of the few systems where these conditions are relatively easy to encounter is represented by beams in bending vibration.

3 Bending vibration of beams

For a more thorough explanation of the principles of vibration in mechanical systems, refer to Sections 1.5.3 and 2.4.2 of [2] and Chapter 2 of [3]. For an in-depth explanation of the vibration of beams, refer to Sections 8.1 to 8.5 of [3].

3.1 Natural frequency and mode shape

In the case of a beam undergoing bending vibration, Equation (2) takes the following form:

$$-\frac{\partial^2}{\partial x^2} \left[EI(x) \frac{\partial^2 y(x, t)}{\partial t^2} \right] = \rho(x) \frac{\partial^2 y(x, t)}{\partial t^2}, \quad (4)$$

where E is the Young's modulus (a parameter representing the elasticity of the material), $I(x)$ is the cross-sectional moment of inertia around the z-axis and $\rho(x)$ is the linear density of the beam (i.e. the mass per unit of length). It can be noticed how none of these parameters depends on an external force, but only on the beam itself.

The solution $y(x, t)$ of (4) can be divided into a spatial component $Y(x)$ and a time component $C(t)$. $Y(x)$ represents the beam profile and coincides with the mode shapes, while $C(t)$ determines the change of the displacement amplitude with time.

$$y(x, t) = Y(x)C(t). \quad (5)$$

Equation (4) can be divided into two different equations, one as a function of $Y(x)$ and the other of $C(t)$. It can be demonstrated that the solution to the time-domain equation is:

$$C(t) = C_0 \cos(\omega t - \phi), \quad (6)$$

where C_0 is the amplitude, ω the frequency of oscillation in *rad/s* and ϕ the phase. The space-domain equation takes the form shown in (7):

$$\frac{d^2}{dx^2} \left[EI(x) \frac{d^2 Y(x)}{dx^2} \right] = \omega^2 \rho(x) Y(x). \quad (7)$$

If the beam is uniform, both $I(x)$ and $\rho(x)$ are constant and all the parameters can be regrouped in the constant β^4 . Equation (7) can then be rewritten as:

$$\frac{d^4 Y(x)}{dx^4} - \beta^4 Y(x) = 0, \quad \beta^4 = \frac{\rho \omega^2}{EI}. \quad (8)$$

The solution of (8) can be verified to be:

$$Y(x) = A \sin(\beta x) + B \cos(\beta x) + C \sinh(\beta x) + D \cosh(\beta x). \quad (9)$$

Example: mode shape of a cantilever beam

To retrieve the values of the parameters A , B , C , D and β , boundary conditions are needed. Let's consider the case of the cantilever beam in Figure (1).

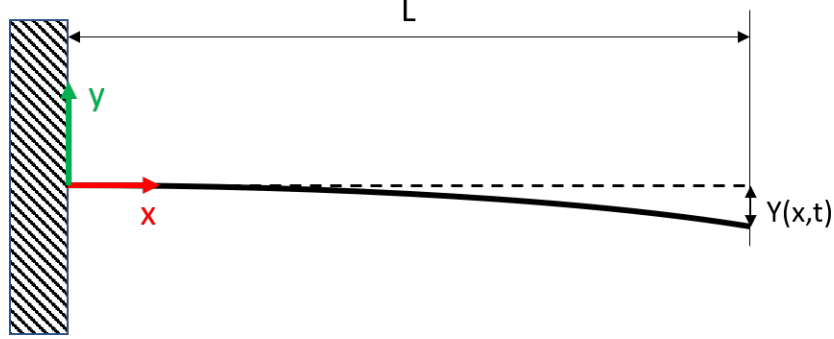


Figure 1: Cantilever beam

In this case, the beam is clamped at one end and free at the other. At the clamped end, the displacement and the first derivatives are zero, yielding:

$$Y(x) = 0, \quad EI \frac{dY(x)}{dx} = 0, \quad x = 0. \quad (10)$$

At the free end of the cantilever, both the bending moment $M(L)$ and shearing forces $Q(L)$ are zero, yielding:

$$M(x) = EI \frac{d^2Y(x)}{dx^2} = 0, \quad Q(x) = -\frac{d}{dx} \left[EI \frac{d^2Y(x)}{dx^2} \right] = 0, \quad x = L. \quad (11)$$

Applying (9) to (10) yields:

$$Y(0) = B + D = 0 \rightarrow D = -B, \quad (12)$$

$$\left. \frac{dY(x)}{dx} \right|_{x=0} = \beta(A + C) = 0 \rightarrow C = -A. \quad (13)$$

Entering the values of C and D in (9) yields:

$$Y(x) = A(\sin(\beta x) - \sinh(\beta x)) + B(\cos(\beta x) - \cosh(\beta x)). \quad (14)$$

Applying (14) to (11) yields:

$$M(L) = -\beta^2 EI [A(\sin(\beta L) + \sinh(\beta L)) + B(\cos(\beta L) + \cosh(\beta L))] = 0, \quad (15)$$

$$Q(L) = \beta^3 EI [A(\cos(\beta L) + \cosh(\beta L)) - B(\sin(\beta L) - \sinh(\beta L))] = 0. \quad (16)$$

Equations (15) and (16) yield:

$$B = -\frac{\sin(\beta L) + \sinh(\beta L)}{\cos(\beta L) + \cosh(\beta L)} A, \quad (17)$$

$$\cos(\beta L) \cosh(\beta L) = -1. \quad (18)$$

Equation (18) has an infinite number of solutions and it can only be solved numerically. Its first four solutions s_1 to s_4 are listed in Table (3.1).

$s_1 = \beta_1 L$	1.88
$s_2 = \beta_2 L$	4.69
$s_3 = \beta_3 L$	7.85
$s_4 = \beta_4 L$	10.995

Table 1: Solutions of equation (18)

Entering the obtained values of s_r in (8) yields the values of all the resonance frequencies of the cantilever beam.

$$\omega_r = s_r^2 \sqrt{\frac{EI}{\rho L^4}}, \quad r = 1, 2, 3 \dots \quad (19)$$

The expression of all the mode shapes of the cantilever is given in (20):

$$Y_r(x) = A \left[\sin(\beta_r x) - \sinh(\beta_r x) - \frac{\sin(\beta_r L) + \sinh(\beta_r L)}{\cos(\beta_r L) + \cosh(\beta_r L)} (\cos(\beta_r x) - \cosh(\beta_r x)) \right]. \quad (20)$$

The first two mode shapes of the cantilever beam are shown in Figure 2.

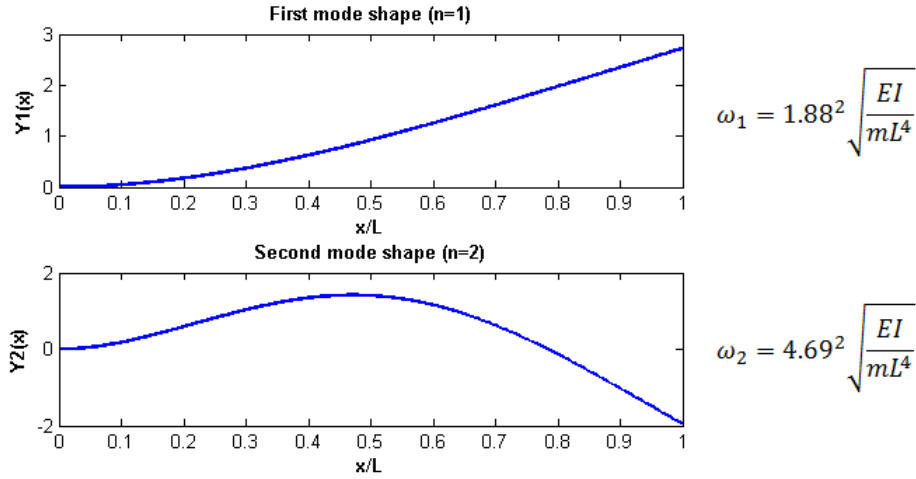


Figure 2: First and second mode shape of a cantilever beam

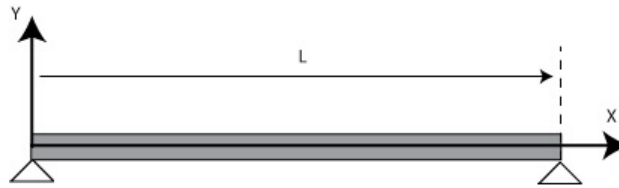


Figure 3: Simply supported beam

It must be noticed how the mode shapes and resonance frequencies change depending on the boundary conditions. Consider the case of a simply supported beam (the supports take forces in x and y direction but do not take bending moments), like the one in Figure (3). Its boundary conditions are:

$$Y(x) = 0, \quad M(x) = EI \frac{d^2 Y(x)}{dx^2} = 0, \quad x = 0, \quad (21)$$

$$Y(x) = 0, \quad M(x) = EI \frac{d^2 Y(x)}{dx^2} = 0, \quad x = L. \quad (22)$$

By applying these conditions to (9), we will obtain different values of ω_r and different mode shapes with respect to the case of the cantilever beam. Still, this does not come as a contradiction to what is explained in Section 2, since the boundary conditions do not depend on any external force and directly influence the stiffness of the beam itself.

3.2 Transfer function and operating deflection shape

To calculate the operating mode shapes, we have to consider the external force $f(x, t)$ acting on the system. Equation (4) has to be modified in this regard to obtain the following expression:

$$\frac{\partial^4 y(x, t)}{\partial x^2} + \frac{\rho}{EI} \frac{\partial y^2(x, t)}{\partial t^2} = \frac{f(x, t)}{EI}. \quad (23)$$

The solution is, in this case, more complicated than in the case of free vibration, and explaining the solving process would go beyond the purpose of this document. For these reasons, the transfer function relating the output displacement to the force is shown directly in Equation (24).

$$G(s) = \frac{y(x_s, s)}{f(x_a, s)} = \sum_{n=1}^{\infty} c_n \frac{Y_n(x_s)Y_n(x_a)}{s^2 + \omega_n^2}, \quad (24)$$

where x_a is the position of the actuator, x_s the position of the sensor and c_n is just a coefficient (see [4]). This equation shows that the number of poles is infinite. More importantly, it also shows that when the position x_a of the actuator or x_s of the sensor coincides with a node (i.e. a point where the mode shape is zero), the transfer function equals zero and the mode shape is *not observable*.

If an accelerometer is used instead of the position sensor, the corresponding transfer function is given by (25).

$$H(s) = s^2 G(s) = \sum_{n=1}^{\infty} s^2 c_n \frac{Y_n(x_s)Y_n(x_a)}{s^2 + \omega_n^2}. \quad (25)$$

4 Exercise Instruction

4.1 Preparation (Homework)

- Using the values of $\beta_r L$ from Table 3.1 and entering them in Equation (20), draw the second, third and fourth mode shapes of a cantilever beam. Consider $A = 1$
- Only for the second and third mode shape:* obtain the position of the nodes, i.e. the values of normalized length (i.e. x/L) for which the value of the mode shape $Y_r(x)$ becomes zero

4.2 Precautions



Failing to respect the following warnings may cause serious damage to the equipment and/or to the setup!

- The reference voltage to actuate a cantilever should be $5V_{op}$ or less; otherwise, the actuator might burn.**
- Use $\pm 10V$ power supply for the setup. Make sure that both channels of the power supply are connected correctly and turned on**

4.3 Experiments

Before starting the experiments, connect the oscilloscope and the cantilever setup as shown in Fig. 4.

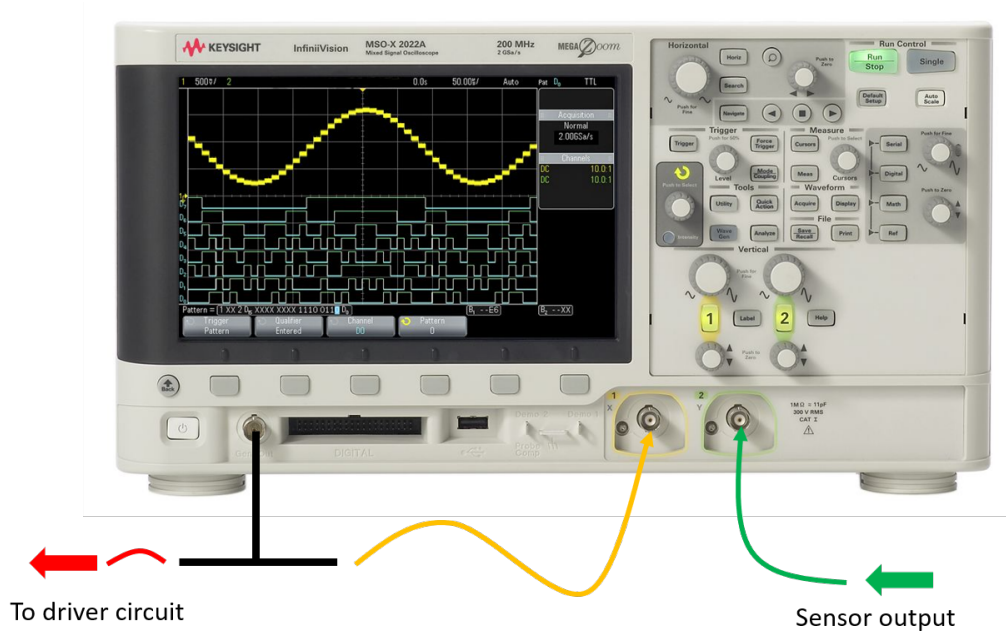


Figure 4: Schematic of the connection plan.

- Position the actuator on the cantilever a few centimetres above the ground and the accelerometer at the top. Let the cantilever oscillate by using the wave generator of the oscilloscope. Find the second and third natural frequencies by adjusting the frequency of the generated wave manually.

2. Trace a Bode plot from 1 Hz to 100 Hz and sketch a pole-zero map, assuming that all poles and zeros are on the imaginary axis.
3. Position the accelerometer on a node of the third resonance frequency. Then, draw a Bode plot and pole-zero map from the second resonance frequency to 100 Hz.
4. Position the sensor several centimetres below the node of experiment 2. Then, trace a Bode plot from the second resonance frequency to 100 Hz and sketch a pole-zero map.
5. Position the sensor several centimetres above the node of experiment 2. Then, trace a Bode plot from the second resonance frequency to 100 Hz and sketch a pole-zero map.
6. Position the accelerometer at the top of the cantilever. By measuring the settling time of the cantilever's swing at the second natural frequency, calculate the time constant and Q factor corresponding to this frequency.

4.4 Hints and Recommendations

1. Use the USB outlets in the lab desk for the oscilloscope and your PC.
2. Use the given MATLAB script for creating the Bode plot ("start.m"). You might have to enter the ID of your oscilloscope. You can get the ID in NI MAX under Devices and Interfaces (Geräte und Schnittstellen).
3. If you abort the Bode plot measurement prematurely (Ctrl + C), the corresponding measurement object was not closed properly. To fix this, you can either restart MATLAB or run the command "instrreset" to close ALL measurement connections.

4.5 Report

Write a report that includes the preparation and experiment results. Add the following points:

- From experiments 2, 3 and 4, explain what happens to the zeros and poles when the third mode is not observable.
- Explain how the sensor and actuator positions are selected in Experiment 6.
- Explain how the dynamics of the system can be described via the state-space representation. By using the state-space representation, explain how the observability and controllability of the states of the system are influenced by the position of the sensor and actuator.

References

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